Carnegie Mellon University

12-771

Applied Data Analytics for Civil and Environmental Systems

# Semester Project

# Regional Characteristics of the Chesapeake Bay Watershed

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# 1 Introduction

The Chesapeake Bay is the largest estuary in North America, with a drainage basin that covers more than 64,000 square miles of land. As a biodiversity hot spot hosting over 3,600 species, the Chesapeake Bay Watershed (CBW) also provides drinking water for 18 million residents and supports large seafood and tourism industries [5]. These ecological, economic, and human health impacts underscore the importance of watershed conservation.

Stormwater runoff, "one of the most harmful sources of pollution to the Bay" [6], presents a significant challenge to preservation efforts. As runoff travels through the watershed, it collects contaminants—primarily nitrogen, phosphorus, and sediment from agricultural and urban sources before discharging into the Bay. These pollutants trigger harmful algal blooms creating low-oxygen "dead zones" and block sunlight essential for aquatic plants and animals [11]. These environmental effects also directly impact fishing, tourism, and residents' welfare [5].

Climate change is projected to intensify the historically increasing precipitation in the Mid-Atlantic region, generating greater runoff and pollutant loading. Adaptation strategies include Best Management Practices (BMPs) designed to improve stormwater quality before discharge.

My Ph.D. research evaluates BMP removal efficiencies for nutrients and sediment under various climate change scenarios in the CBW using APEX modeling. Given it's spatial diversity, my efforts focus four physiographic regions within the watershed: Appalachia, Ridge and Valley, Piedmont, and Coastal Plain. These four provinces represent distinct landscapes within the watershed with varying geologic formations, topographical features, and hydrological characteristics. To accurately represent these regions, I chose to analyze three key characteristics: soil type, slope, and groundwater depth; using open-source data to determine representative values for modeling inputs and better understand their differences.

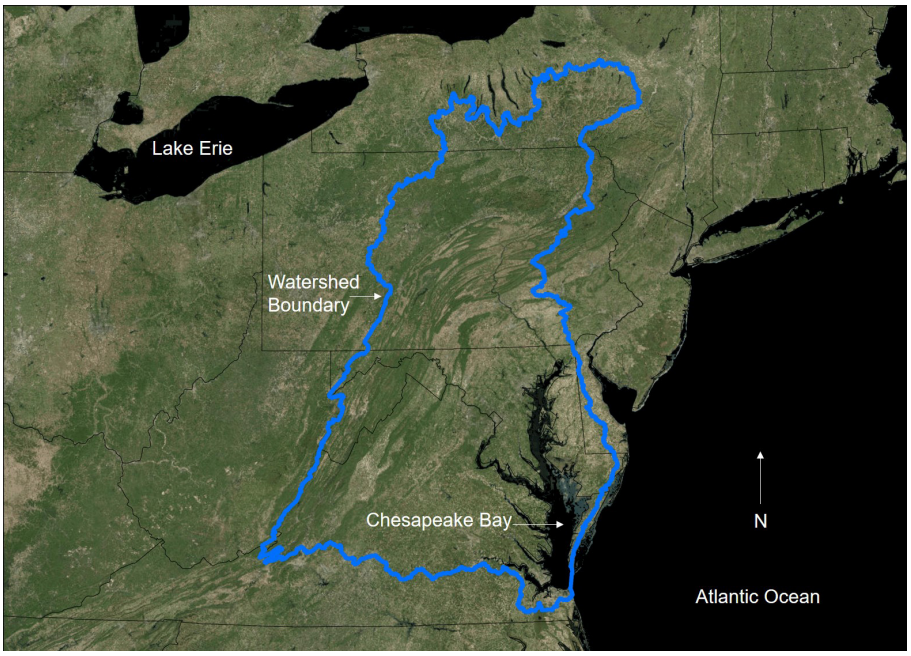


Figure 1: Chesapeake Bay Watershed Boundary, via US Geologic Survey

# 2 Methods

Soil type, slope, and groundwater depth were chosen as defining characteristics for the physiographic regions of the CBW. Mean values for each region were estimated using various analytical methods to aggregate open-source datasets. Shape files were prepared delineating the boundaries of the four physiographic regions of interest: Appalachia, Coastal Plain, Piedmont, and Ridge and Valley.

Kruskal-Wallis tests were performed using NumPy and SciPy to evaluate whether statistically significant differences existed between soil types and groundwater depths across the four physiographic regions. This nonparametric approach was selected because it does not assume normal distribution of the data and is appropriate for comparing multiple independent groups with different sample sizes. Statistical analyses were not deemed appropriate for the slope data due to sample size limitations. A significance level of 0.05 was used for both tests.

## 2.1 Slope

### Data Selection and Preparation

To calculate the mean slope for each region, void-filled data from the Shuttle Radar Topography Mission (SRTM) digital elevation model (DEM) with a resolution of 30 meters (1 arc-second) and vertical accuracy of +/- 16 meters [9]. The resolution was deemed sufficiently detailed to capture topographic variation on the regional-scale used while maintaining computational efficiency. Therefore, SRTM was chosen for its optimal balance between spatial coverage and resolution, and accuracy for regional-scale terrain analysis.

Before starting the analysis, regional boundary shape files for each physiographic region were uploaded to Google Earth Engine as project assets.

The analysis was implemented through Google Earth Engine (GEE) using JavaScript programming to avoid local computational limitations associated with managing and then "mosaicking" multiple large DEM files, as well as projection distortion issues introduced by coordinate system transformations. GEE's cloud-based infrastructure provided efficient processing capabilities for the large-scale terrain analysis required for each region.

### DEM Processing and Slope Calculation

To begin DEM processing, the SRTM dataset was accessed through the GEE catalog (USGS/SRTMGL1_003) using the ee.Image() constructor to allow for access through the GEE API. Then, the GEE terrain.slope() function was used to calculate the percent slope (rise/run * 100) for each 30-meter pixel. This applied a slope raster to the entire dataset.

For each physiographic province, the slope raster was clipped to the regional boundary using GEE's mask function. This allowed for the extraction of slope values within each region, excluding outside areas. Zonal statistics were calculated using the GEE reduceRegion() function with a mean reducer to derive the average slope percentage across all pixels in each clipped region. The scale was set to 30 meters to maintain SRTM's native resolution and a maxPixels parameter was set to 1e9 to accommodate regions with large amounts of pixels.

### Post-Processing

The final processing was performed using Python's Pandas and GeoPandas libraries to add the geometries of each region to the slope data. Pandas was used to read the GEE analysis output as a DataFrame before adding the geometries extracted from each regional boundary shapefile. The dataframe was then converted to a GeoPandas GeoDataFrame and plotted using the Matplotlib library, creating a map that displays the mean slope values alongside physiographic boundaries.

## 2.2 Soils

### Data Selection and Preparation

The gridded National Soil Survey Geographic Database (gNATSGO) was used for representative soil analysis courtesy of USDA-NRCS Soil & Plant Science Division (SPSD) [12]. Downloaded as an ESRI file geodatabase, it is a composite of data from the Soil Survey Geographic Database (SSURGO), State Soil Geographic Database (STATSGO2), and Raster Soil Survey Databases (RSS). It was selected because it provides the most comprehensive national soil data at a resolution of 30 m, which is appropriate for regional analysis given the scale of each province. The database contains raster data with soil type identifiers (MUKEYs) accompanied by attribute tables with detailed soil characteristics.

Python's rasterio library was used for raster data processing. The raster containing soil map units (MUKEY raster) was extracted from the gNATSGO geopackage and then clipped to each regional boundary shapefile, while preserving the original 30-meter resolution to maintain spatial detail.

### Soil Classification

For each region, the most common soil type (MUKEY) was identified based on pixel count. Python's NumPy library was used to tally pixel values, find the most frequent MUKEY, and calculate the associated coverage as a percentage.

The initial MUKEY-based analysis proved too granular for regional characterization, prompting a conceptual shift from taxonomic to functional classification by focusing on hydrologic soil groups (HSGs). HSGs are more relevant in the context of hydrologic modeling as they directly relate to runoff generation and infiltration rates (Table 1). In addition, higher- classification makes regional patterns more apparent and facilitates comparison across provinces.

Table 1: Hydrologic Soil Group (HSG) Classifications

| Group | Description |
|---|---|
| A | Well-drained (high infiltration) |
| B | Moderately well-drained (moderate infiltration) |
| C | Poorly drained (slow infiltration) |
| D | Very poorly drained (very slow infiltration) |
| A/D | Dual: A when drained, D when natural |
| B/D | Dual: B when drained, D when natural |
| C/D | Dual: C when drained, D when natural |

Thus, the previous NumPy approach was expanded to create a dictionary of all unique MUKEYs and their pixel counts within each region. MUKEYs, which are numeric codes, were then used to retrieve the corresponding soil properties from attribute tables. The SQLite3 module in Python was used to connect to the gNATSGO database, allowing for the interpretation of the spatial data. A systemic exploration of the database structure was necessary to identify exactly where and how to extract HSG information. SQL queries were executed to examine the available tables and inspect their contents; this yielded the discovery of the "component" table containing the "hydgrp" column. Further sample queries were run to confirm that the "hydgrp" field contained HSG classifications. With this information, another SQL query was conducted, targeting the component table to extract the MUKEY, HSG, and component percentage needed to calculate the HSG distribution in each region.

#### HSG Distribution Calculation

To calculate the HSG distribution for each region, a two- weighted approach was developed to account for (1) component-level variation: individual MUKEYs often have multiple components with different HSGs; and (2) spatial distribution: different MUKEYs cover varying proportions of each region. Thus, the percentage of each HSG in each region was determined by multiplying the component percentage, or fraction of the MUKEY represented by the component, by the percentage of the region's area covered by the MUKEY, and then summing the contributions across all regional MUKEYs.

#### Post-Processing

A dictionary was created to map HSG codes to detailed descriptions and associated regional percentages were organized into a DataFrame form using Python's Pandas library. Matplotlib was used to generate visualizations of the data. Pie charts were selected to visualize proportional distributions for individual regions and a stacked bar chart was created to facilitate cross-region comparison.

### 2.3 Groundwater Depth

#### Data Selection and Preparation

Groundwater depth data was accessed thorugh the U.S. Geological Survey (USGS)'s National Water Information System (NWIS) [13]. Discrete measurements, which are manual, field-based depth-to-water readings in wells, were selected instead of continuous monitoring data due to broader spatial coverage (35,689 discrete measurement sites in Pennsylvania compared to only 364 continuously monitored stations. This approach was deemed appropriate because this analysis aimed to determine a single representative value for each region rather than to analyze temporal patterns.

To manage computational resources and maintain the scope of the project, the analysis focused on representative counties within each physiographic region (see Table 2 and Figure 1). These counties were previously selected as part of my hydrologic modeling in APEX based on regional agricultural yields.

Data files containing groundwater depth measurements were stored in tab-delimited text format. Using Python's Pandas module, these data were converted to DataFrames and underwent formatting adjustments to facilitate analyses. Specifically, the data

Table 2: Representative Counties for Physiographic Regions

| Region | County | No. of Sites |
|---|---|---|
| Appalachia | Steuben, NY | 2 |
| Ridge and Valley | Bedford, PA | 1 |
| Piedmont | Frederick, MA | 5 |
| Coastal Plain | Dorchester, MD | 18 |

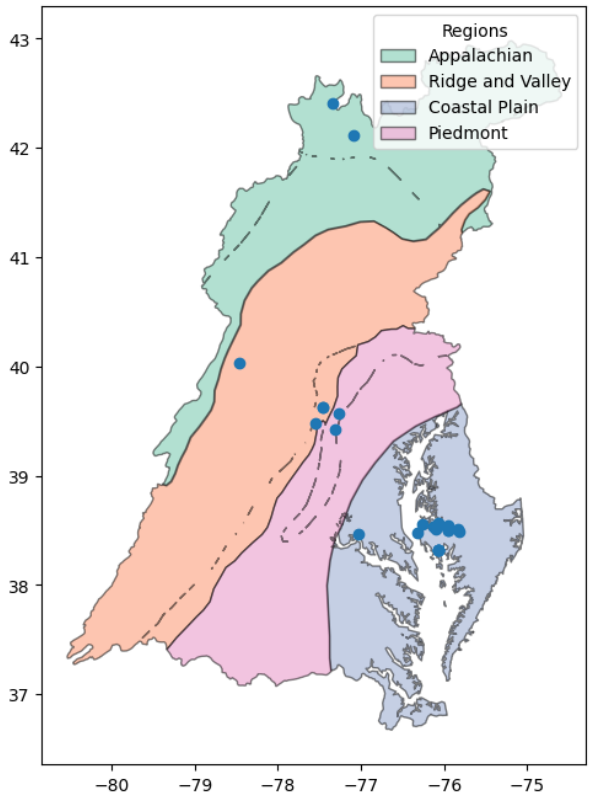


Figure 2: USGS Groundwater Monitoring Sites

were filtered to include only depth-to-water measurements (parameter code 72019), date columns were converted to datetime format, and water level values were converted to numeric type. Erroneous values were handled using Pandas' coercion capabilities and the dropna() function was used to exclude missing values. The monitoring period selected was 2020-2025 and groundwater depths were recorded as feet below land surface.

### Time Series Analysis

To begin the analysis, groundwater depths for each monitoring station were plotted against time and grouped by province to identify regional patterns over the entire monitoring period.

For a seasonal analysis, monthly averages were also calculated for each station and averaged across all years to identify potential seasonal fluctuations and how they might vary by region. This process involved extracting the month component from the measurement dates and grouping the data accordingly.

The final visualization considered higher-level regional averages and was achieved by plotting the average groundwater depths of all stations in a given region. Using the

scipy.stats module, a linear regression analysis was performed on the data to identify long-term trends. The resulting regression lines were plotted alongside the time series data. No data transformations were applied prior to regression analysis, as the primary goal was to identify general trends rather than to develop predictive models.

### Mean Groundwater Depth Calculation

To obtain representative groundwater depths for each province, region-specific measurements were first aggregated by date and sorted chronologically before calculating the mean over the complete time series of a region. To minimize potential bias in the regional averaging process, station-specific means were first calculated before aggregating to regional scale to prevent stations with more frequent measurements from dominating the regional average. All monitoring stations within each region were equally weighted.

# 3 Results

## 3.1 Slopes

Representative slopes for each region ranged from about 3-10%. The Coastal Plain and Piedmont regions are characterized by gentler slopes, contrasting moderate slopes in Appalachia and Ridge and Valley. Table 2 shows the average slope in each province.

Table 3: Mean Slope by Physiographic Region

| Region | Slope (%) |
|---|---|
| Appalachia | 8.68 |
| Ridge and Valley | 9.56 |
| Coastal Plain | 3.29 |
| Piedmont | 4.72 |

Figure 2 provides a graphic visualization of the regional slopes, highlighting the contrast between the more mountainous Appalachia and Ridge and Valley with the Piedmont and Coastal Plain closer to the Chesapeake Bay.

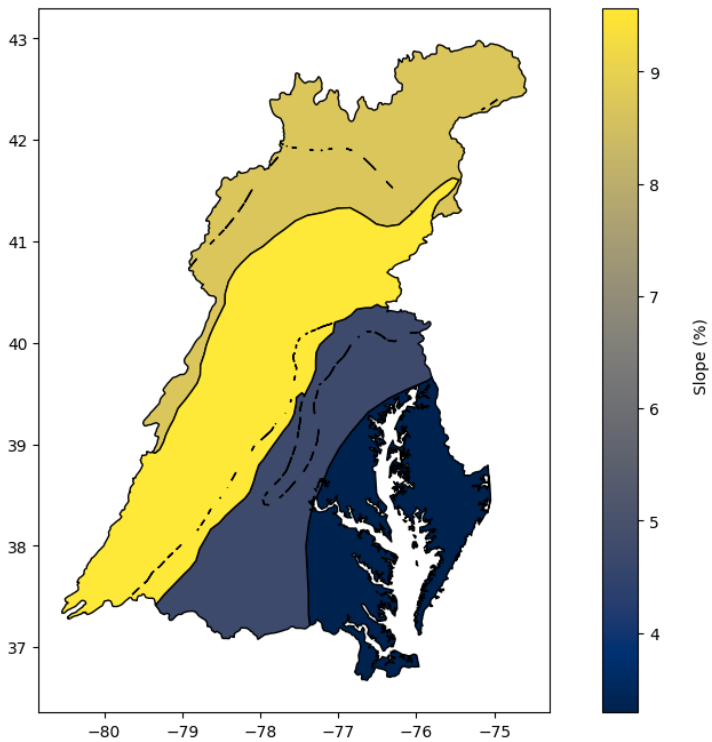


Figure 3: Average Slope by Physiographic Region

## 3.2 Soils

HSG type B (moderately well-drained) was the most common soil group in all regions besides Appalachia, dominated by HSG type D. Although Appalachia, Ridge and Valley, and Coastal Plain share a primary soil type, their percentages vary significantly. Table 3 summarizes the leading HSG in each region.

Table 4: Most Common HSG by Physiographic Region

| Region | HSG | Percentage (%) |
|---|---|---|
| Appalachia | D | 39.3 |
| Ridge and Valley | B | 40.1 |
| Coastal Plain | B | 21.2 |
| Piedmont | B | 55.3 |

Regional HSG distributions are presented as a stacked bar chart for comparison of drainage characteristics between regions in Figure 3.

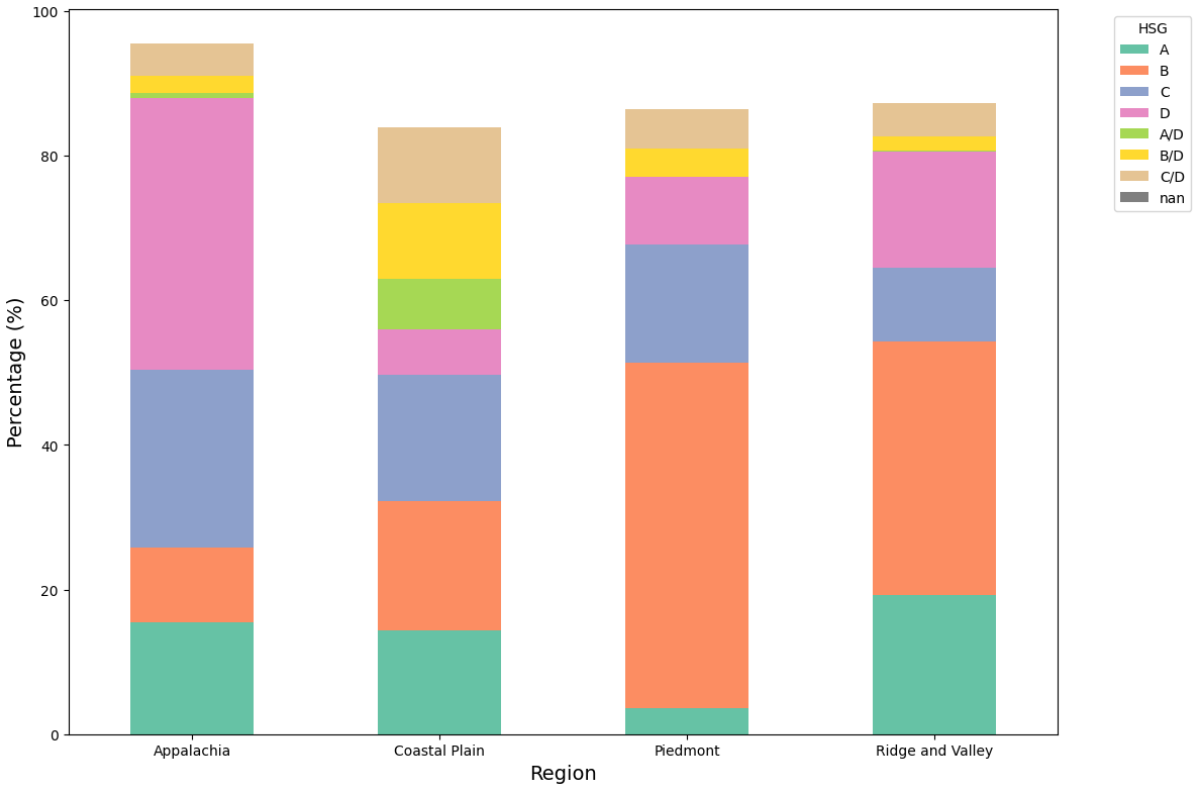


Figure 4: Comparative Analysis of HSGs Across Physiographic Regions

The bar chart highlights Appalachia's distinct poorly drained profile and predominantly moderate drainage conditions in the Piedmont region. Soil distributions appear to be more complex in the Coastal Plain and Ridge and Valley. Figure 4 presents a more detailed visualization of each region's soil composition.

Appalachia is mainly made up of very poorly drained (39.3%) and poorly drained (25.8%) soils, which reflects the region's mountainous topography. Steep slopes accelerate runoff, thus limiting water infiltration and promoting erosion [2]. The region's extensive mining history is another factor since coal mining inverted soil profiles, burying fertile topsoil beneath less permeable subsoil layers [3].

Soils in the Ridge and Valley region are mostly moderately well-drained (40.1%) to well-drained (22%). The region's characteristic alternating steep slopes promote runoff and valleys with deep soils facilitate infiltration. Deeper bedrock in the valley regions also allow for more drainage [7].

The distribution of soil types in the coastal plain follows a more balanced pattern,

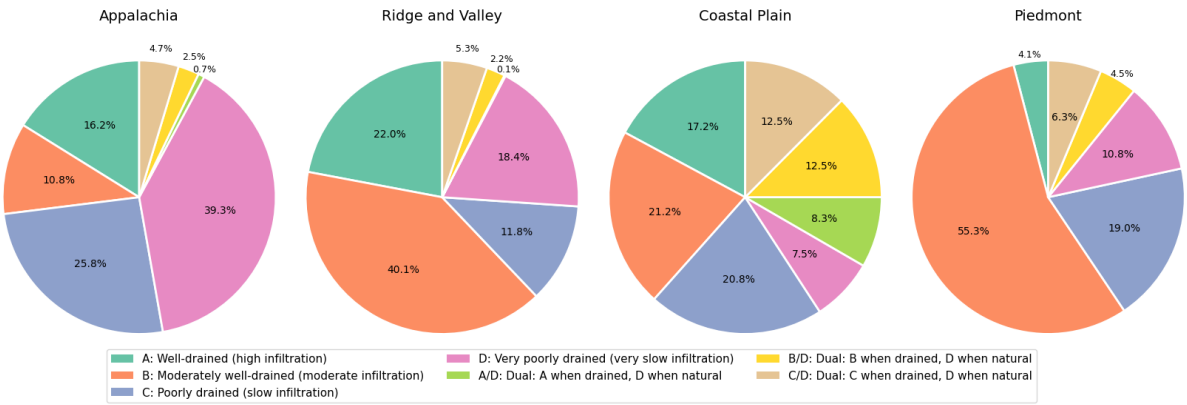


Figure 5: Distribution of HSGs by Physiographic Region

reflecting more variability. The estuarine characteristics of the region yield complex soil profiles composed of multiple sediment layers from various deposit sources (i.e., beach, fluvial, offshore). Flat topography and high water tables contribute to poor drainage, while sandier soils are associated with good infiltration [4].

The piedmont region is overwhelmingly dominated by moderately well-drained soils (55.3%). A contributing factor is soil makeup, with infiltration being facilitated by sandy topsoils, before being limited by clayey subsoils [8].

### Statistical Analysis

The Kruskal-Wallis test revealed no significant differences (H=0.79, p-value=0.85) in soil distribution between regions. The high p-value indicates that the variability in the values is likely consistent with random variation. This suggests that HSGs are relatively homogeneously distributed throughout the watershed area.

## 3.3 Groundwater Depth

Mean groundwater depths ranged from about 12-40ft. Table 5 provides a summary of the average depth for each region.

Table 5: Mean Depth to Groundwater by Physiographic Region

| Region | Depth (ft) |
|---|---|
| Appalachia | 40.8 |
| Ridge and Valley | 16.3 |
| Coastal Plain | 12.4 |
| Piedmont | 36.1 |

With a mean groundwater depth of 40.8ft, Appalachia appears to have the deepest water table. Steep slopes and higher elevations are correlated with deeper water tables [4], which is the case for this mountainous region. The Coastal Plain has the shallowest mean groundwater depth (12.4ft), likely due to its low-lying topography and proximity the ocean among other effects.

### Temporal Analysis

Figure 5 displays average groundwater depth measurements across all sites within each region during the five year monitoring period, each point representing a regional average for a specific date. Regression lines representing the best-fit linear relationship between groundwater depth and time are also included to highlight long-term regional behavior.

The low correlation coefficient, R, for each region (ranging from 0.02 to 0.15) reveal a poor linear fit, suggesting that temporal trends do not have a significant effect on the overall variability in groundwater depth. However, each region's regression line has a constant positive slope, pointing to gradually decreasing water tables as the depth to groundwater increases. This is consistent with the current state of groundwater resources within the watershed mostly due to over-extraction and reduced recharge rates [10]. Therefore, the regressions provide information on directional trends despite their low correlation.

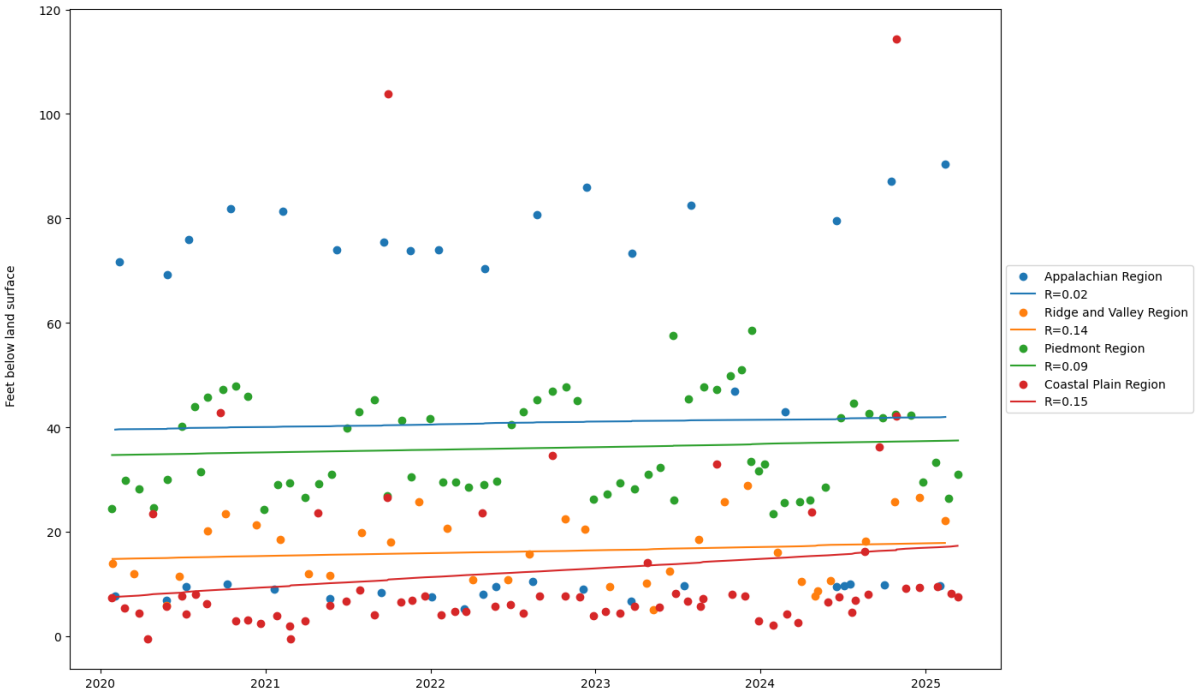


Figure 6: Regional Groundwater Depth with Linear Regression Analysis (2020-2025)

The depths of groundwater in the Coastal Plain exhibit the strongest increasing trend (R=0.15) at the steepest rate, while Appalachia and Piedmont have the lowest slope with the weakest correlations (R=0.02 and R=0.09, respectively).

### Seasonal Analysis

Seasonal variation is represented in Figure 6, which shows monthly patterns averaged over the five year period, where each color represents a different USGS monitoring station.

Patterns on a monthly scale vary significantly between stations in the Appalachian and Piedmont regions. Stations with deeper mean depths (blue) in the Appalachian region exhibit more seasonal variation than shallower wells (orange). This appears to be the opposite in the Piedmont region, where deeper wells (blue) have relatively constant groundwater depth measurements, while shallower sites (red and purple) show a clear trend of deeper groundwater tables in the summer and fall months.

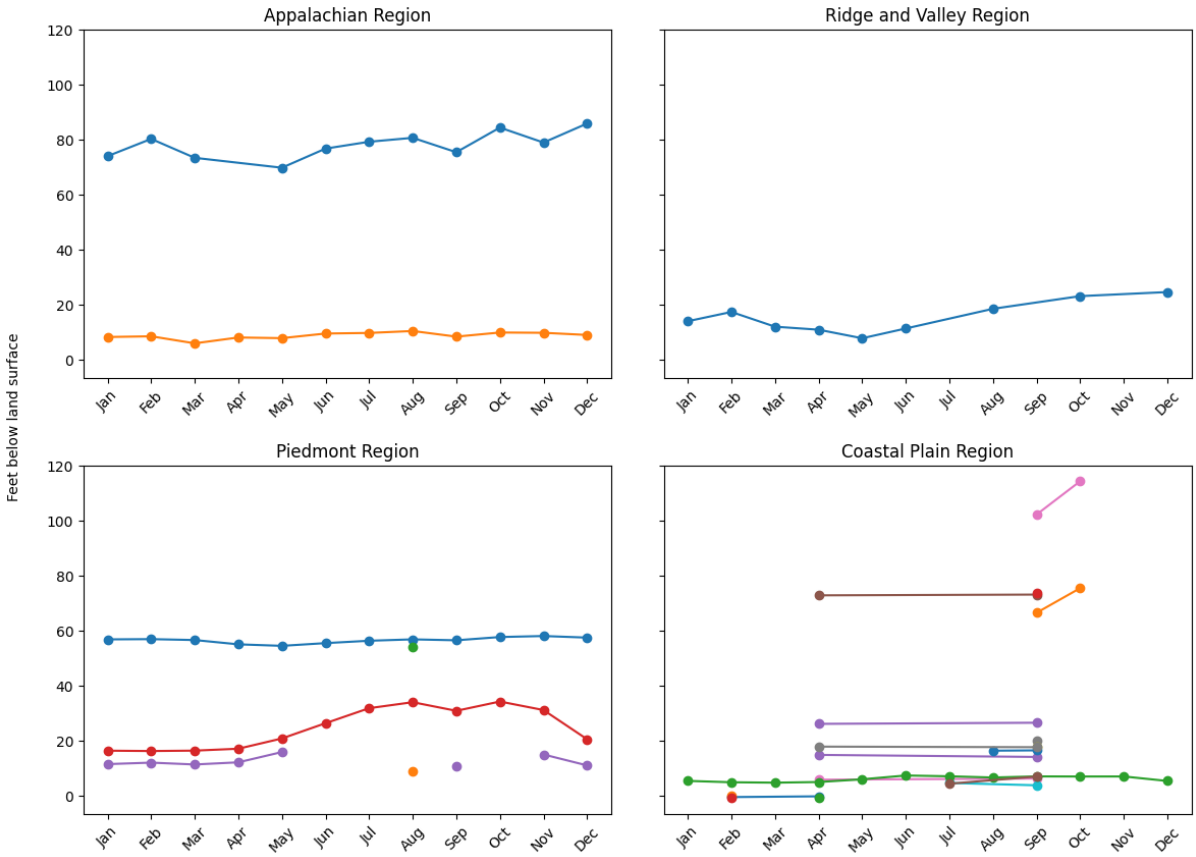


Figure 7: Monthly Groundwater Depth by Physiographic Region (Averaged Over 2020-2025)

There is only one monitoring site in the Ridge and Valley region, making it difficult to compare between different well depths. However, the one site presents a trend similar to the Piedmont site at a similar mean depth (red), with increasing groundwater depths beginning in May. Contrasting the Piedmont region, Ridge and Valley groundwater depths follow an increasing trend beyond the fall and into the winter months.

The Coastal Plain appeared to have the least seasonal variability, with relatively constant groundwater depths throughout all months for each station.

### Individual Monitoring Sites

All groundwater depth measurements for each station within each region are displayed in Figure 7.

This highlights the variation between the different stations. Within each region, stations mean depth and trends vary significantly. The Piedmont and Ridge and Valley regions have more stations displaying cyclical patterns, whereas the Coastal Plain and Appalachia are more consistent. The Coastal Plain also appears to show the most inter-site variability with several distinct depth ranges; it is also the region with the largest number of monitoring sites.

Figure 8: Groundwater Depths at Individual Monitoring Sites by Physiographic Region (2020-2025)

### Statistical Analysis

The Kruskal-Wallis test revealed statistically significant differences (H = 100.1, p-value = 1.47e-21) in groundwater depth between the regions. The extremely low p-value indicates high confidence in this outcome.

# 4 Discussion

The scale of the CBW was especially apparent when identifying representative characteristics for each region. Although I understood its size conceptually, exploring data for these characteristics helped me better grasp the true magnitude of the watershed's complexity. Each physiographic region has such an extensive and diverse terrain that determining single representative values for slope, soil, and groundwater depth sometimes felt like an oversimplification. For instance, I was particularly struck by how soil diversity manifested within provinces – even the most prevalent soil types made up only a small percentage of a region's total soil composition.

Outside of the representative values themselves, statistical analyses revealed patterns regarding how soil and groundwater depth vary by region. Groundwater depth showed statistically significant differences between regions, while soil was more uniform with three of the four regions sharing a predominant soil type. This could be partially attributed to the broadness of the HSG classification system, compared to depth measurements. In addition, soil formation processes are more sensitive to external climate, while the groundwater table dynamics are governed primarily by geology [1]. Climate is more homogeneous throughout the watershed than geology, which varies greatly by province, which could contribute to more regional water table variation.

Future work could involve post-hoc analyses for both soils and groundwater to examine specific pairwise regional differences, expanding upon the Kruskal-Wallis results. Additional value could come from cross-characteristic comparisons to identify potential relationships between these factors within regions.

Groundwater analyses could also be expanded upon. This includes further trend analyses at different time steps, as well as a deeper examination of monitoring station variations. Incorporating aquifer mapping could help further understand regional differences and the influence of potential vertical layering effects.

Finally, integrating land use distribution analyses would also enhance understanding of each regional profile and how they differ from each other. It could also provide explanations for the some of the patterns observed in the other characteristics.

# 5 Conclusion

An analysis of representative topography, soil type, and groundwater depth was performed for the Appalachia, Ridge and Valley, Piedmont, and Coastal Plain regions of the CBW through primarily geospatial analytical techniques. This analysis has contributed to a better understanding of the regional variety in the watershed and allowed reasonable inputs to be identified for representative hydrological modeling of the area. The findings show that Appalachia had a mean slope of 8.68% and a mean groundwater depth of 40.8 ft, the deepest among the regions. It was also the only region dominated by very poorly drained soils (HSG D), while the other three had mostly moderately well-drained soils (HSG B). Ridge and Valley had the steepest average slope at 9.56% and a mean groundwater depth of 16.3ft. The Piedmont region had intermediary values with mean slope and depth to groundwater of 4.72% and 36.1ft, respectively, and the Coastal Plain had the lowest average slope and shallowest water table (3.29% and 12.4ft, respectively). Kruskal-Wallis tests revealed that groundwater depths are significantly different between regions (H=100, p-value=1.47e-21) while soil types are not (H=0.787, p-value=0.853). These inputs will be directly used for model building using APEX to observe the effects of climate change on best management practices and inform decision making regarding climate-adaptive conservation planning for each province of the watershed.

# A Appendices

Appendices A.1-A.3 contain the code used for the analyses. Plotting code is only included in Appendix A.3 (groundwater depth calculations) because the analysis and visualization components are closely integrated in this section.

## A.1 Slope Analysis

### A.1.1 Google Earth Engine Implementation (Java)

```
// load SRTM data
var srtm = ee.Image('USGS/SRTMGL1_003');

// calculate slope
var slope = ee.Terrain.slope(srtm);

// define regions
var regionAssets = [
  'projects/chesapeake-watershed-analysis/assets/appalachia_shp_dis',
  'projects/chesapeake-watershed-analysis/assets/coastal_shp_dis',
  'projects/chesapeake-watershed-analysis/assets/piedmont_shp_dis',
  'projects/chesapeake-watershed-analysis/assets/valley_shp_dis'
];

// array to store results
var results = [];

// calc slope for each region
regionAssets.forEach(function(assetId) {
    // extract region name from asset ID
    var regionName = assetId.split('/').pop();

    // load region
    var region = ee.FeatureCollection(assetId);

    // calculate mean slope
    var meanSlope = slope.reduceRegion({
      reducer: ee.Reducer.mean(),
      geometry: region.geometry(),
      scale: 30,
      maxPixels: 1e9
});

    // get result
    var meanSlopeValue = meanSlope.get('slope');

    // add to results
    results.push({
```

```
      'region': regionName,
      'mean_slope_percent': meanSlopeValue
    });

// print all results
print('Results:', results);

// prepare results for export
var resultsCollection = ee.FeatureCollection(
  results.map(function(result) {
    return ee.Feature(null, result);
  })
);
```

### A.1.2 Python Post-Processing

```
# load df with slope results from Google Earth Engine analysis
region_slopes_df = pd.read_csv(region_slopes_fp)

# define regions
region_key = {'appalachia_shp_dis': 'Appalachia',
              'coastal_shp_dis': 'Coastal Plain',
              'piedmont_shp_dis': 'Piedmont',
              'valley_shp_dis': 'Ridge and Valley'}

# add region names to df
region_slopes_df['region_name'] = region_slopes_df['region'].map(
    region_key)

# add actual geometries of each region to data

geometries = {} # to store geometries by region

# find all .shp files
region_folders = [f for f in os.listdir(regions_fp) if os.path.
    isdir(os.path.join(regions_fp, f))]

# process each region
for region_folder in region_folders:
    region_path = os.path.join(regions_fp, region_folder,f"{
        region_folder}.shp")

    # read shapefile
    region_gdf = gpd.read_file(region_path)

    # add to geometries dict
    geometries[region_folder] = region_gdf.geometry.iloc[0]

print(geometries)
# add to df
```

```python
region_slopes_df['.geo'] = region_slopes_df['region'].map(geometries)

# convert to GeoDataFrame
region_slopes_gdf = gpd.GeoDataFrame(region_slopes_df, geometry='.geo', crs=region_gdf.crs)

# plot gdf
```

## A.2 Soil Analysis (Python)

### A.2.1 Soil Identification Key (MUKEY) Retrieval

```python
output_dir = '/Users/mayastruzak/Documents/Academic/12771_Project/clipped-soil'

# open national raster
src_main = rasterio.open(gnatsgo_fp+raster_fn)

# empty list to store results
results = []

# find all .shp files
region_folders = [f for f in os.listdir(regions_fp) if os.path.isdir(os.path.join(regions_fp, f))]

# process each region
for region_folder in region_folders:
    region_path = os.path.join(regions_fp, region_folder)

    # get shp file
    shp_files = glob.glob(os.path.join(region_path, "*.shp"))

    if shp_files:
        shp_file = shp_files[0]
        region_name = os.path.basename(region_folder)

        # load shp
        region_gdf = gpd.read_file(shp_file)

        # match crs
        region_gdf_proj = region_gdf.to_crs(src_main.crs)

        # clip to shapefile
        clipped_band, clip_transform = mask(src_main, region_gdf_proj['geometry'], crop=True)

        # export clipped band
        out_raster = os.path.join(output_dir, f"{region_name}_soil.tif")
        out_meta = src_main.meta.copy()
```

```python
        out_meta.update({
            "driver": "GTiff",
            "height": clipped_band.shape[1],
            "width": clipped_band.shape[2],
            "transform": clip_transform
        })

        # write clipped band
        with rasterio.open(out_raster, 'w', **out_meta) as dest:
            dest.write(clipped_band)

        # analyze clipped band to get representative soil type
        clipped_data = clipped_band[0]

        # count occurance
        valid_data = clipped_data[clipped_data != src_main.nodata]
        if len(valid_data) > 0:
            values, counts = np.unique(valid_data, return_counts=True)

            # get most common soil
            most_common_idx = np.argmax(counts)
            mukey = int(values[most_common_idx])
            area_percentage = (counts[most_common_idx] / np.sum(counts)) * 100

            # save results
            results.append({
                'region': region_name,
                'soil_id': mukey,
                'pixel_count': int(counts[most_common_idx]),
                'total_pixels': len(valid_data),
                'percentage': area_percentage
            })


# print results df
soils_df = pd.DataFrame(results)
print(soils_df)

src_main.close()
```

```
            region  soil_id  pixel_count  total_pixels  percentage
0  appalachia_shp   296754       597817      47744515    1.252117
1      valley_shp   545487       381675      64590783    0.590912
2     coastal_shp   118088       278905      33429373    0.834311
3    piedmont_shp   542826       327790      41817672    0.783855
```

### A.2.2 Corresponding Soil Type Identification

```
# connect to the GeoPackage as a SQLite database
conn = sqlite3.connect(gnatsgo_fp+"/gNATSGO_02_03_2025.gpkg")

# query for each mukey in results
for soil in soils_df['soil_id']:
    soil_query = f"""
    SELECT *
    FROM component
    WHERE mukey = {soil}
    """
    soil_data = pd.read_sql_query(soil_query, conn)

    # save results to CSV
    soil_data.to_csv(f"/Users/mayastruzak/Documents/Academic
       /12771_Project/soil_{soil}_data.csv", index=False)

# close connection
conn.close()
```

### A.2.3 HSG Retrieval

```
# connect to the GeoPackage as a SQLite database
conn = sqlite3.connect(gnatsgo_fp+"/gNATSGO_02_03_2025.gpkg")

region_hsg = {}

# process each raster
for region in soils_df['region']:
    # open region-clipped raster
    with rasterio.open(f"{output_dir}/{region}_soil.tif") as src:
        # get unique MUKEYs and counts
        soil_data = src.read(1)
        values, counts = np.unique(soil_data[soil_data != src.
           nodata], return_counts=True)
        mukey_counts = dict(zip(values,counts))
        total_pixels = sum(counts)

        # get HSG data
        query = f"""
        SELECT mukey, hydgrp, comppct_r
        FROM component
        WHERE mukey IN ({','.join(map(str, values))})
        """
        hsg_data = pd.read_sql_query(query, conn)

        # calulctae HSG percentages
        hsg_pct = {}
        for hsg in hsg_data['hydgrp'].unique():
            hsg_pct[hsg] = 0
```

```python

            # sum weighted percentages
            for _, row in hsg_data[hsg_data['hydgrp'] == hsg].iterrows():
                mukey = row['mukey']
                if mukey in mukey_counts:
                    hsg_pct[hsg] += (row['comppct_r'] / 100) * (mukey_counts[mukey] / total_pixels) * 100

        region_hsg[region] = hsg_pct

# save results to df
hsg_results = []
for region, hsgs in region_hsg.items():
    for hsg, pct in hsgs.items():
        hsg_results.append({
            'region': region,
            'hsg': hsg,
            'percentage': round(pct,3)
        })
hsg_df = pd.DataFrame(hsg_results)

conn.close()

# store HSG % for each rgeion
region_data = {}

# collect HSG % values
for region in hsg_df['region'].unique():
    region_data[region] = hsg_df[hsg_df['region'] == region]['percentage'].values
```

### A.2.4 Kruskal-Wallis Test

$H_o$ = distribution of mean groundwater depths are the same across all regions
$H_A$ = distribution of mean groundwater depths vary between regions

```python
h_stat, p_value = kruskal(*region_data.values())
```

## A.3 Groundwater Depth Analysis

### A.3.1 Data Processing

```python
# define file paths and rows to skip in csv
files_info = [
    (ap_gwl_fp, 84),
    (rv_gwl_fp, 86),
    (pd_gwl_fp, 92),
    (cp_gwl_fp, 102)
]

```

```python
# dictionary to store dataframes
gwl_dfs = {}

# extract base names for dictionary keys
file_names = ['ap_gwl', 'rv_gwl', 'pd_gwl', 'cp_gwl']

# get header with column names
with open(files_info[0][0], 'r') as f:
    lines = f.readlines()
    header_line = lines[83].strip()
header_parts = header_line.split()

# loop through files and process each
for i, (file_path, skiprows) in enumerate(files_info):
    # Read data
    df = pd.read_csv(file_path, sep='\t', skiprows=skiprows, names=header_parts)
    df = df.drop(0)
    #df = df.dropna()  # Keeping this commented as in your original code
    df.reset_index(drop=True, inplace=True)

    # Store dataframe in dictionary
    gwl_dfs[file_names[i]] = df

# set up the region_dfs
region_dfs = {
    'Appalachian': gwl_dfs['ap_gwl'],
    'Ridge and Valley': gwl_dfs['rv_gwl'],
    'Piedmont': gwl_dfs['pd_gwl'],
    'Coastal Plain': gwl_dfs['cp_gwl']
}

# filter and format data
for region, df in region_dfs.items():
    # filter to depth to wl
    region_dfs[region] = df[df['parameter_cd'] == '72019']
    # convert columns to appropriate types
    region_dfs[region]['lev_dt'] = pd.to_datetime(region_dfs[region]['lev_dt'])
    region_dfs[region]['lev_va'] = pd.to_numeric(region_dfs[region]['lev_va'], errors='coerce')
```

### A.3.2 Mean Depth Calculation

```python
region_means = {}
for region, df in region_dfs.items():
    # Group by date and avg across all sites in the region
    region_avg = df.groupby('lev_dt')['lev_va'].mean().reset_index()
    region_avg = region_avg.sort_values('lev_dt')
```

```python

    # Remove any NaN values
    region_avg = region_avg.dropna(subset=['lev_va'])

    # Calculate mean
    mean_value = region_avg['lev_va'].mean()

    # Store in dictionaryt
    region_means[region] = mean_value
print(region_means)
```

### A.3.3 Monitoring Sites

```python
# geographic data for each station (NAD83)
sta_info = {
    'USGS_Site_No': [
        '420710077052101', '422445077203301', '400217078281901', '392517077190401',
        '392834077331401', '393423077154701', '393733077274801', '393738077275101',
        '381911076042901', '381917076042801', '381919076043201', '381920076042401',
        '382800076180701', '382847076190901', '382916075491702', '382945075570201',
        '382945075570203', '383056075500201', '383100076061103', '383151076080801',
        '383225075565002', '383243076042801', '383256076035301', '383328076153602',
        '383340076041601', '383408076042402'
    ],
    'Latitude': [
        42.11955556, 42.4124722, 40.0376111, 39.42149125, 39.47627778, 39.5725861,
        39.62593114, 39.6272222, 38.319681, 38.32131944, 38.3218611, 38.32229,
        38.4666, 38.48019444, 38.48808, 38.49575, 38.49575, 38.51555556,
        38.51666667, 38.5308333, 38.54102778, 38.5453968, 38.5488611, 38.55789866,
        38.5617222, 38.56905556
    ],
    'Longitude': [
        -77.0890556, -77.34227778, -78.4719167, -77.3174838, -77.5539083, -77.26297778,
        -77.4630433, -77.4641667, -76.074597, -76.0744, -76.07553889, -76.07341,
        -77.0365, -76.31938889, -75.8214, -75.9505833, -75.9505833, -75.83388889,
        -76.1033333, -76.1352222, -75.9461111, -76.0741067, -76.0642222, -76.259668,
        -76.0705556, -76.07288889
```

```
    ]
}

# create the dataframe
field_sta_df = pd.DataFrame(sta_info)

# convert to geodataframe
geometry = gpd.points_from_xy(field_sta_df['Longitude'], field_sta_df['Latitude'])
field_sta_gdf = gpd.GeoDataFrame(field_sta_df, geometry=geometry, crs='4269')

# Plot
fig, ax = plt.subplots(1,1, squeeze=False, sharex=True, sharey=True, gridspec_kw={'wspace':0.5, 'hspace':0.5})


# legend support: handles and labels
handles = []
labels = []
region_aliases = {'appalachia_shp': 'Appalachian',
                  'valley_shp': 'Ridge and Valley',
                  'piedmont_shp': 'Piedmont',
                  'coastal_shp': 'Coastal Plain'}

for i, (region_name, gdf) in enumerate(region_gdf.items()):
    gdf = region_gdf[region_name]
    color = plt.cm.Set2(i)
    # use plot polygon for legend support
    gdf.plot(ax=ax[0,0], color=color, alpha=0.5, edgecolor='black')

    # patch for legend
    patch = plt.Rectangle((0,0), 1, 1, color=color, alpha=0.5, ec='black')
    handles.append(patch)

    alias = region_aliases.get(region_name, region_name)
    labels.append(alias)

ax[0,0].legend(handles, labels,loc='best', title='Regions')

# plot stations
field_sta_gdf.plot(ax=ax[0,0])

# ax[0,0].set_title('USGS Groundwater Monitoring Stations')

fig.set_size_inches(12, 8)
plt.show()
```

### A.3.4 Field Measurements for All Stations by Region

```
# plot all GW depth data
fig, axs = plt.subplots(2,2, squeeze=True, sharex=False, sharey=True, gridspec_kw={'wspace':0.1, 'hspace':0.25})
axs = axs.flatten()

for i, (region, df) in enumerate(region_dfs.items()):
    ax = axs[i]
    sites = df['site_no'].unique()

    # plot each site with a different color
    for site in sites:
        site_data = df[df['site_no'] == site]
        ax.scatter(site_data['lev_dt'], site_data['lev_va'], label=site)

    ax.set_title(f'{region} Region')
    # ax.legend(title='Site No.',loc='center left', bbox_to_anchor=(1, 0.5), fontsize=10)

fig.text(0.075, 0.5, 'Feet below land surface', va='center', rotation='vertical')
fig.tight_layout()
# fig.text(0.5, 0.95, 'Groundwater Levels - Field Measurements', ha='center', va='bottom', size=16)
fig.set_size_inches(15,10)
# plt.subplots_adjust(right=1)

plt.show()
```

### A.3.5 Monthly Averages of All Stations by Region

```
fig, axs = plt.subplots(2, 2, squeeze=True, sharex=False, sharey=True, gridspec_kw={'wspace':0.1, 'hspace':0.25})
axs = axs.flatten()

for i, (region, df) in enumerate(region_dfs.items()):
    ax = axs[i]
    sites = df['site_no'].unique()

    # plot each site with a different color
    for site in sites:
        site_data = df[df['site_no'] == site]

        # extract just the month from the dates
        site_data['month'] = site_data['lev_dt'].dt.month

        # group by month only, averaging across all years
        monthly_avg = site_data.groupby('month')['lev_va'].mean().reset_index()
```

```python

        # sort data by month
        monthly_avg = monthly_avg.sort_values('month')

        # get month names for x-axis
        month_names = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec']
        monthly_avg['month_name'] = monthly_avg['month'].apply(lambda x: month_names[x-1])

        # plot monthly averages
        ax.plot(monthly_avg['month'], monthly_avg['lev_va'], marker='o', linestyle='-', label=site)

    # Set x-axis ticks and labels
    ax.set_xticks(range(1, 13))
    ax.set_xticklabels(month_names)
    plt.setp(ax.xaxis.get_majorticklabels(), rotation=45)

    ax.set_title(f'{region} Region')
    # ax.legend(title='Site No.', loc='center left', bbox_to_anchor=(1, 0.5), fontsize=10)

fig.text(0.075, 0.5, 'Feet below land surface', va='center', rotation='vertical')
# fig.text(0.5, 0.95, 'Average Groundwater Levels by Month (All Years)', ha='center', va='bottom', size=16)
fig.set_size_inches(15, 10)
plt.subplots_adjust(right=0.85)
plt.show()
```

### A.3.6 Regional Averages with Linear Regression

```python
fig, ax = plt.subplots(1, 1, figsize=(15, 10))

for region, df in region_dfs.items():
    # Group by date and avg across all sites in the region
    region_avg = df.groupby('lev_dt')['lev_va'].mean().reset_index()
    region_avg = region_avg.sort_values('lev_dt')

    # remove nan values
    region_avg = region_avg.dropna(subset=['lev_va'])

    # plot data pts
    ax.scatter(region_avg['lev_dt'], region_avg['lev_va'], label=f'{region} Region')


    # calculate regression
    x = np.arange(len(region_avg))
```

```
    y = region_avg['lev_va'].values
    result = scipy.stats.linregress(x, y)
    y_pred = result.slope * x + result.intercept

    ax.plot(region_avg['lev_dt'], y_pred, label=f'R={result.rvalue:.2f}')

# ax.set_title('Average Groundwater Levels by Region', size=16)
ax.legend(loc='center left', bbox_to_anchor=(1, 0.5), fontsize=10)
fig.text(0.075, 0.5, 'Feet below land surface', va='center', rotation='vertical')
plt.subplots_adjust(right=0.85)
plt.show()
```

### A.3.7 Kruskal-Wallis Test

$H_o$ = distribution of mean groundwater depths are the same across all regions
$H_A$ = distribution of mean groundwater depths vary between regions

```
region_values = {}
for region, df in region_dfs.items():
    region_avg = df.groupby('lev_dt')['lev_va'].mean().reset_index()
    region_avg = region_avg.dropna(subset=['lev_va'])
    region_values[region] = region_avg['lev_va'].values

print(scipy.stats.kruskal(*[region_values[region] for region in region_values.keys()]))
```

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